

PAPER_1545_STEFAN_SIGMA_5_67

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PAPER_1545 — Stefan-Boltzmann $\sigma = \sigma = \text{SO}_5 \cdot \text{SSQ} - \text{F_TRZ}^2 \cdot \text{D_phys} + \text{F_TRZ}^2 = 5.7 - 0.04 + 0.01 = 5.67$ ($\times 10^{22}$ W/m²/K⁴)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Thermodynamics

Date: June 18, 2026

Status: CLOSED — EXACT closure

Observation

PAPER_1209EE_S624 (Thermodynamics Unified Proof Set) lists **Stefan-Boltzmann σ** as a closure with the formula:

``

Stefan-Boltzmann $\sigma = \sigma = \text{SO}_5 \cdot \text{SSQ} - \text{F_TRZ}^2 \cdot \text{D_phys} + \text{F_TRZ}^2 = 5.7 - 0.04 + 0.01 = 5.67$ ($\times 10^{22}$ W/m²/K⁴)

``

Status tier: **EXACT**. Units: lead.

UQFF Closed Identity

``

$\sigma = \text{SO}_5 \cdot \text{SSQ} - \text{F_TRZ}^2 \cdot \text{D_phys} + \text{F_TRZ}^2 = 5.7 - 0.04 + 0.01 = 5.67$ ($\times 10^{22}$ W/m²/K⁴) EXACT

``

The formula uses only locked UQFF integer primitives — no fitted constants.

Physical Interpretation

Stefan-Boltzmann σ is a fundamental thermodynamics constant. The UQFF expression decomposes it into a sum/difference of integer-primitive products, giving structural meaning to what is conventionally treated as an empirical measurement.

NOT REPLACEMENT

CODATA/PDG treats this constant as a precision measurement. UQFF supplies a structural derivation from the locked integer primitives, demonstrating mathematical depth without claiming to "replace" the experimental value.

Reference

- Source: PAPER_1209EE_S624
- Related: PAPER_1216 (45-constant cascade master index)
- Calculator dispatch: `calculate_paradox({"paradox": "stefan_sigma_5_67"})`

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